
RELIABILITY PRODUCT REVIEW

FROM ASSET EVENT TO REUSABLE LEARNING.

A practical review mechanism for selecting, designing, and governing digital reliability use cases.

CANDIDATE THESIS

Use this review before a pilot, at a product increment, after a material exception, and before a scale decision.

REVIEW FLOW

- 01 Event and consequence
- 02 Context and actionability
- 03 Evidence and authority
- 04 Learning and reuse

1 | QUALIFY THE DECISION, NOT THE TECHNOLOGY

CONSEQUENCE	What asset, safety, quality, delivery, or economic outcome is exposed?	Criticality, downtime category, customer/process consequence
RECURRENCE / BURDEN	Is this repeated enough, costly enough, or risky enough to learn from?	Event history, labor/rework, recurring machine or quality issues
ACTIONABILITY	What decision or work step can change when insight arrives?	Decision owner, threshold, response window, work route
DATA READINESS	Can asset, work, condition, failure, and material context be connected?	Completeness, lineage, timing, naming, ownership
ADOPTION BURDEN	What must people do differently?	Workflow steps, cognitive load, training, incentives, exceptions
REUSE	Is the capability portable across sites or asset classes?	Common object model, legitimate variation, integration pattern

2 | MINIMUM RELIABILITY DATA CONTRACT

REQUIRED CONTEXT

- Asset identity, hierarchy, criticality
- Observed condition and timestamp
- Work order, task, labor, closure evidence
- Failure mode, mechanism, cause, uncertainty
- Material, substitute, lead time, availability
- Decision, owner, date, expected effect

GOVERNANCE THAT CHANGES WORK

- Named owner for each reusable data product
- Quality checks embedded in workflow
- Controlled vocabulary with exception path
- Discoverable lineage and consumption
- Feedback when data is insufficient
- Retirement rules for unused capture

3 | BOUND AI BY WORK AND AUTHORITY

DETECT	Meaningful anomaly with sufficient signal and context.	Confirm relevance; reject noise.
PRIORITIZE	Consequence, urgency, capacity, and dependencies can be compared.	Approve queue and tradeoffs.
RECOMMEND	Options, rationale, uncertainty, and evidence are visible.	Select, modify, defer, or escalate.
EXPLAIN	Pattern and source evidence can be traced.	Validate against domain knowledge.
AUTOMATE	Bounded low-risk action, tested controls, fallback, owner, and monitoring.	Set policy, monitor exceptions, halt.

4 | MAKE THE PRODUCT DECISION

SCALE

Observed value, usable workflow, governed evidence, ownership, reusable architecture.

ADAPT

Core capability transfers; asset, site, or process variation is legitimate and bounded.

HOLD

Consequence may be real, but actionability, data, ownership, or adoption is not ready.

BUY / CONFIGURE

Capability is commodity and fits architecture without surrendering the decision model.

BUILD

Workflow, integration, data product, or decision experience is differentiating.

STOP

The use case cannot produce enough value or learning to justify its burden.

5 | CLOSE THE LEARNING LOOP

Record what changed in the work, what evidence improved, what users accepted or overrode, what outcome was observed, what should be standardized, and what should remain local. A pilot that leaves no reusable asset is only a demonstration.

ASSUMPTION DISCIPLINE

No score or percentage is assigned without a Carpenter baseline. Each field is a review question, not a judgment about current maturity.